ՀԱՅԱՍՏԱՆԻ ՀԱՆՐԱՊԵՏՈՒԹՅՈՒՆ ԿՐԹՈՒԹՅԱՆ ԵՎ ԳԻՏՈՒԹՅԱՆ ՆԱԽԱՐԱՐՈՒԹՅՈՒՆ
ՀԱՅԱՍՏԱՆԻ ԱԶԳԱՅԻՆ ՊՈԼԻՏԵԽՆԻԿԱԿԱՆ ՀԱՄԱԼՍԱՐԱՆ
ՏՀՏԷ ԻՆՍՏԻՏՈՒՏ ՏԵՂԵԿԱՏՎՈՒԹՅԱՆ ԱՆՎՏԱՆԳՈՒԹՅԱՆ ԵՎ ԾՐԱԳՐԱՅԻՆ ԱՊԱՀՈՎՄԱՆ ԱՄԲԻՈՆ

Լաբորատոր աշխատանք 1

Ինստիտուտ՝ ՏՀՏԷ
Առարկա՝ Ծրագրային ապահովման անվտանգություն
Թեմա՝ Լիցենզիայի ծածկագրեր և դրանց կապը ապարատային ապահովման հետ
Խումբ ՝ SS-355
Ուսանող՝ Մելքոնյան Սոնա

Երևան 2026

Կոնֆիգուրացիա VS 2026 + x64 համար
cmake -G "Visual Studio 18 2026" -A x64 -S .. -B .
Build անել Debug կոնֆիգուրացիայով
cmake --build . --config Debug

```
x64 Native Tools Command P

C:\Users\user\Desktop\Software Security\Lab3>cd "C:\Users\user\Desktop\Software Security\Lab3\build_vs"

C:\Users\user\Desktop\Software Security\Lab3\build_vs>cmake -G "Visual Studio 18 2026" -A x64 -S .. -B .
-- Selecting Windows SDK version 10.0.26100.0 to target Windows 10.0.26200.
-- The C compiler identification is MSVC 19.50.35727.0
-- The CXX compiler identification is MSVC 19.50.35727.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: C:/Program Files/Microsoft Visual Studio/18/Community/VC/Tools/MSVC/14.50.35717/bin/Hostx64/x64/cl.exe - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: C:/Program Files/Microsoft Visual Studio/18/Community/VC/Tools/MSVC/14.50.35717/bin/Hostx64/x64/cl.exe - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (6.9s)
-- Generating done (0.0s)
-- Build files have been written to: C:/Users/user/Desktop/Software Security/Lab3/build_vs

C:\Users\user\Desktop\Software Security\Lab3\build_vs>cmake --build . --config Debug
Версия MSBuild 18.4.0+6e61e96ac для .NET Framework

  1>Checking Build System
  Building Custom Rule C:/Users/user/Desktop/Software Security/Lab3/CMakeLists.txt
  main.cpp
  HardwareID.cpp
C:\Users\user\Desktop\Software Security\Lab3\src\HardwareID.cpp(11,9): warning C4005: WIN32_LEAN_AND_MEAN: изменение ма
кроопределения [C:\Users\user\Desktop\Software Security\Lab3\build_vs\LicensingSystem.vcxproj]
      C:\Users\user\Desktop\Software Security\Lab3\src\HardwareID.cpp(11,9):
      "WIN32_LEAN_AND_MEAN" ранее объявлен в командной строке

  LicenseKey.cpp
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseKey.cpp(34,34): warning C4244: =: преобразование "time_t" в "un
signed long", возможна потеря данных [C:\Users\user\Desktop\Software Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseKey.cpp(68,36): warning C4996: 'ctime': This function or variab
le may be unsafe. Consider using ctime_s instead. To disable deprecation, use _CRT_SECURE_NO_WARNINGS. See online help
for details. [C:\Users\user\Desktop\Software Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseKey.cpp(71,41): warning C4996: 'ctime': This function or variab
le may be unsafe. Consider using ctime_s instead. To disable deprecation, use _CRT_SECURE_NO_WARNINGS. See online help
```

```
x64 Native Tools Command P

C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(87,22): warning C4566: символ, представленный униве
рсальным именем символа "\u056C", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u0532", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u0548", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u053C", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u0550", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u053B", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u0551", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u0535", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u0546", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u0536", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u0531", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "\u0538", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
  Создание кода...
  LicensingSystem.vcxproj -> C:\Users\user\Desktop\Software Security\Lab3\build_vs\Debug\LicensingSystem.exe
  Building Custom Rule C:/Users/user/Desktop/Software Security/Lab3/CMakeLists.txt

C:\Users\user\Desktop\Software Security\Lab3\build_vs>
```

Կիրառում ենք մենյույում արդեն առկա գործիքները

```
x64 Native Tools Command P

C:\Users\user\Desktop\Software Security\Lab3\build_vs\Debug>.\LicensingSystem.exe
==============================================
    LICENSE SYSTEM - LABORATORY WORK
==============================================

Current Device ID 0000000076F677BB
CPU ID  49656E6900000020BFCBFBFF000906A4
MAC address 00:15:5d:01:0f:00


--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
0. Exit
Your choice: 1
Enter licensee name: Ss

!!!! PERPETUAL LICENSE CREATED:
====================================
License Key: 40F3F-5BD40-F3F5B
Licensee: Ss
Issue Date: Wed Mar 18 04:37:48 2026
Type: PERPETUAL
====================================

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
0. Exit
Your choice: 4

????? ???????????:
====================================
License Key: 40F3F-5BD40-F3F5B
Licensee: Ss
Issue Date: Wed Mar 18 04:37:48 2026
Type: PERPETUAL
====================================

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
```

5°C Partly cloudy Поиск ENG 4:39 18.03.2026

```
x64 Native Tools Command P

Your choice: 3
Enter license key: 40F3F-5BD40-F3F5B
v LICENSE IS VALID:

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
0. Exit
Your choice: 2
Enter licensee name: AMrr

v TEMPORARY LICENSE CREATED (30 days):
====================================
License Key: 7447A-3F874-47A3F
Licensee: AMrr
Issue Date: Wed Mar 18 04:38:40 2026
Valid until: Fri Apr 17 04:38:40 2026
Type: TEMPORARY
====================================

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
0. Exit
Your choice: 4

????? ???????????:
====================================
License Key: 40F3F-5BD40-F3F5B
Licensee: Ss
Issue Date: Wed Mar 18 04:37:48 2026
Type: PERPETUAL
====================================
====================================
License Key: 7447A-3F874-47A3F
Licensee: AMrr
Issue Date: Wed Mar 18 04:38:40 2026
Valid until: Fri Apr 17 04:38:40 2026
Type: TEMPORARY
====================================

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
```

Rainy days ahead 5°C Поиск ENG 4:41 18.03.2026

Ավելացնում ենք 6-րդ կետը, որպեսզի հնարավորություն ունենանք փոփոխել ժամանակավոր բանալիների տևողության օրերը

main.cpp

```
#include <iostream>#include <string>#include "LicenseManager.h"int main() {
    std::cout << "==================================================" << std::endl;
    std::cout << "    LICENSE SYSTEM - LABORATORY WORK    " << std::endl;
    std::cout << "==================================================" << std::endl;
    std::cout << std::endl;
    LicenseManager manager;
    int choice;
    std::string userName;
    int tempLicenseDays = 30; // Default 30 days for temporary licensesdo {
        std::cout << "\n--- SELECT ACTION ---" << std::endl;
        std::cout << "1. Create perpetual license" << std::endl;
        std::cout << "2. Create temporary license (" << tempLicenseDays << " days)" << std::endl;
        std::cout << "3. Verify license" << std::endl;
        std::cout << "4. Show all licenses" << std::endl;
        std::cout << "5. Save license to file" << std::endl;
        std::cout << "6. Change temporary license duration (current: " << tempLicenseDays << " days)" << std::endl;
        std::cout << "0. Exit" << std::endl;
        std::cout << "Your choice: ";
        std::cin >> choice;
        std::cin.ignore();
        switch (choice) {
        case 1: {
            std::cout << "Enter licensee name: ";
            std::getline(std::cin, userName);

            LicenseKey license = manager.createPerpetualLicense(userName);
            std::cout << "\n!!!! PERPETUAL LICENSE CREATED:" << std::endl;
            license.printInfo();
            break;
        }

        case 2: {
            std::cout << "Enter licensee name: ";
            std::getline(std::cin, userName);

            LicenseKey license = manager.createTemporaryLicense(userName, tempLicenseDays);
            std::cout << "\n!!!! TEMPORARY LICENSE CREATED (" << tempLicenseDays << " days):" << std::endl;
            license.printInfo();
            break;
        }
        case 3: {
            std::string key;
            std::cout << "Enter license key: ";
            std::getline(std::cin, key);

            if (manager.verifyLicense(key)) {
                std::cout << "v LICENSE IS VALID" << std::endl;
            }
            else {
                std::cout << "x LICENSE IS INVALID OR NOT FOUND" << std::endl;
            }
            break; }
 case 4: {
 manager.listAllLicenses();
 break; }
```

```
 case 5: {
        std::string key, filename;
        std::cout << "Enter license key: ";
        std::getline(std::cin, key);
        std::cout << "Enter filename: ";
        std::getline(std::cin, filename);
        LicenseKey temp(manager.getCurrentMachineID(), "Test User");
        if (manager.saveToFile(temp, filename)) {
           std::cout << "v License saved to " << filename << std::endl;}
        else {
           std::cout << "x Error saving to file" << std::endl;
        }
        break;      }
 case 6: {
        std::cout << "Enter new duration in days for temporary license: ";
        int newDays;
        std::cin >> newDays;
        std::cin.ignore();
        if (newDays > 0) {
           tempLicenseDays = newDays;
           std::cout << "Temporary license duration updated to " << tempLicenseDays << " days." << std::endl;
        }
        else {
           std::cout << "Invalid number of days." << std::endl;}
        break;}
 case 0: {
        std::cout << "Program finished." << std::endl;
        break;}
      default: {
        std::cout << "Invalid choice" << std::endl;
        break;}
      }
  } while (choice != 0);
  return 0;}
```

```
C:\Users\user\Desktop\Softw

Type: TEMPORARY
====================================

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
6. Change temporary license duration (current: 30 days)
0. Exit
Your choice: 6
Enter new duration in days for temporary license: 6
Temporary license duration updated to 6 days.

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (6 days)
3. Verify license
4. Show all licenses
5. Save license to file
6. Change temporary license duration (current: 6 days)
0. Exit
Your choice: 4

????? ???????????:
====================================
License Key: 05260-68205-26068
Licensee: VSS
Issue Date: Wed Mar 18 11:52:12 2026
Valid until: Fri Apr 17 11:52:12 2026
Type: TEMPORARY
====================================

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (6 days)
3. Verify license
4. Show all licenses
5. Save license to file
6. Change temporary license duration (current: 6 days)
```